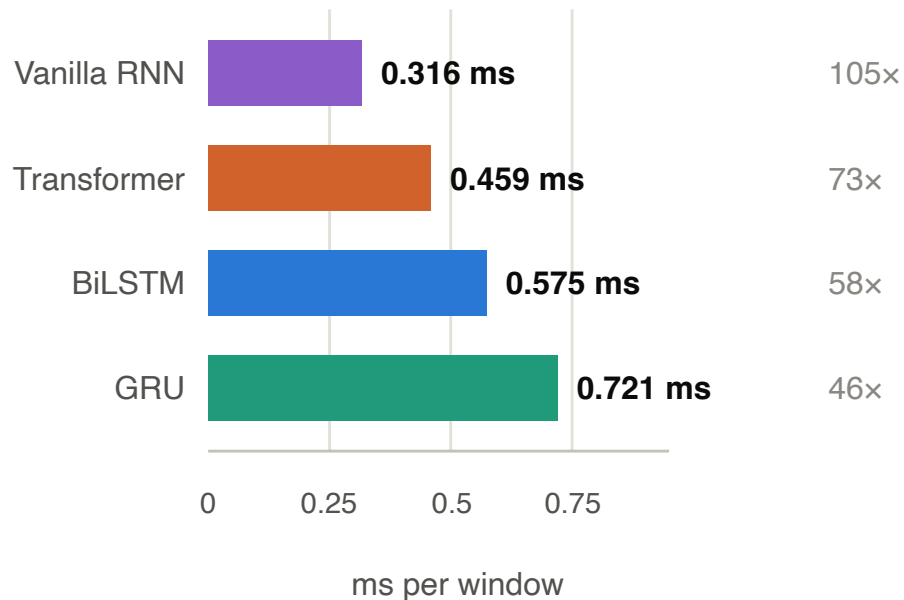


(a) Cost of one prediction

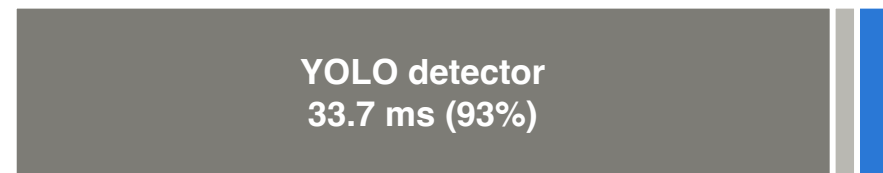
Apple M4 CPU, batch 1, 50 warm-up + 1,000 timed forwards.



× = how many predictions fit in one 33.3 ms frame budget

(b) Where a frame's time actually goes

The same machine, live pipeline. The answer to (a) is irrelevant next to this.



total 36.3 ms per frame

- YOLO detector 33.700 ms 92.7%
- ByteTrack association 1.000 ms 2.8%
- Crossing-intention model 1.647 ms 4.5%

The intention model is 4.5% of the frame. Making it faster changes nothing; the pipeline is detection-bound.